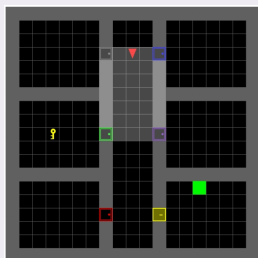


Plan-and-Execute: LLM-Guided Reinforcement Learning with Cross-Modal Fusion for Long-Sequence Decision Making

Yadong Li¹, Tong Zhang¹, Zhen Cui²

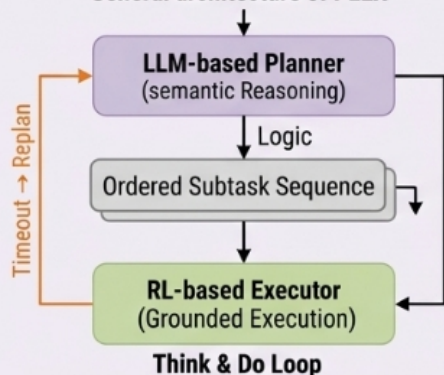
1. Nanjing University of Science and Technology, 2. Beijing Normal University

1. MOTIVATION



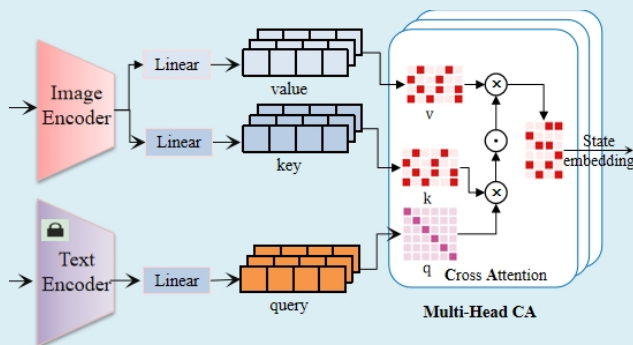
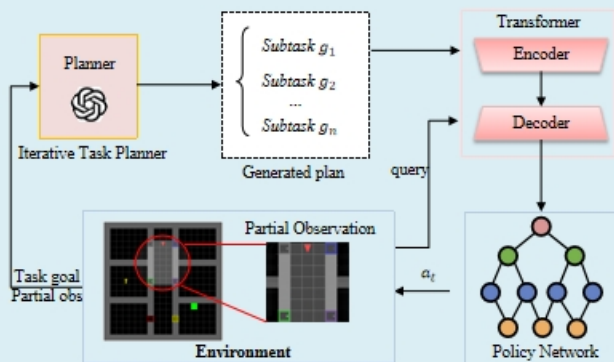
- Sparse Rewards
- Partial observe
- Changed goals

General architecture of PLEX



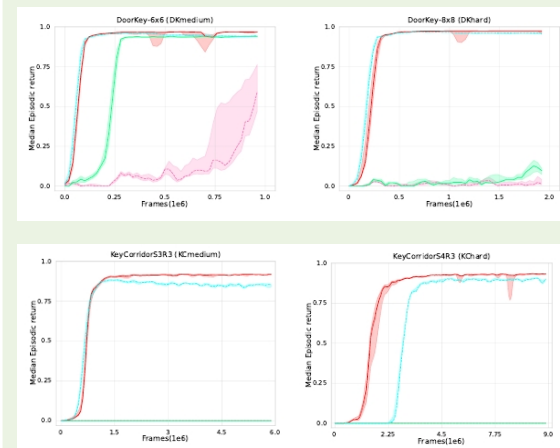
Hierarchical intergration of semantic planning and grounded execution

2. METHOD DETAILS



Language-Grounded Reinforcement Learning

3. EXPERIMENTS



4. CONCLUSION

- Hierarchical LLM-RL enhances long-horizon decision making
- Iterative planning provide robust execution
- Ground policy improve context awareness



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